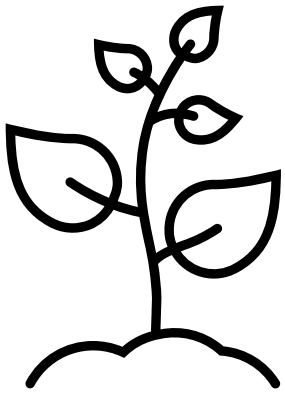
UNIVERSITY of BRADFORD

School of Computer Science, AI & Electronics



Generative Artificial Intelligence Chatbot Using Prompt Engineering

For BNG (Biodiversity Net Gain) Image Processing

Dissertation Project Report

Mishel Rehan

22006867

A thesis submitted in part fulfilment of the degree of BEng (Hons) Software Engineering

Supervisor: Professor Ciprian Daniel Neagu

06 January 2025

Declaration

The candidate confirms that the work submitted is her own and that appropriate credit has been given where reference has been made to the work of others. The candidate agrees that this report can be electronically checked for plagiarism.

Mishel Rehan

Contents

Abstract 1

Acknowledgements 2

1. Chapter 1 3

1.1 Introduction 3

1.1.1 Biodiversity and BNG 3

1.1.2 Generative AI and Image Processing 4

1.1.3 AI's significance in conservation initiatives 4

1.1.4 Prompt Engineering and its significance 4

1.1.5 Existing Gaps in Tools and Methods 5

1.1.6 Importance of AI in Addressing Biodiversity Challenges 5

1.1.7 Challenges in establishing and estimating BNG 7

1.2 Project aims and objectives 8

1.3 Methodology 9

1.4 9

Abstract

Biodiversity Net Gain (BNG), a crucial foundation for promoting ecological diversity and guaranteeing sustainable development, is the mere focus of this project. In order to assist its monitorisation and promotion, this project focuses on the development of a generative AI chatbot which includes an integrated image-processing pipeline. The main goal is to develop a system prototype that can recognise, categorise, and count flora via the analysis of user-uploaded images.

To enhance its comprehension and delivery, the chatbot makes use of OpenAI's GPT-4o, which is a sophisticated generative AI model which been augmented with methods of prompt engineering. Prompt engineering validates precise and context-aware responses, ensuring the chatbot delivers comprehensive evaluations as well as practical insights. The image-processing component utilises cutting-edge deep learning models, which have been trained on the Oxford LibGuides' Plant Taxonomy database, including ResNet for classification and YOLOv5 for object recognition.

Significant project goals are sought to be achieved, such as the building and development of the prototype in order to ensure the smooth process of the chatbot and image processing outputs, as well as the efficient construction of selected models in order to accurately comprehend an assortment of flora. The chatbot aims to assist BNG, rendering improvements in ecological conservation efforts.

Keywords: Biodiversity Net Gain (BNG), Generative AI (GenAI) chatbot, computer vision, image processing, prompt engineering, flora, deep learning.

Acknowledgements

(to be done later)

1. Chapter 1

1.1 Introduction

1.1.1 Biodiversity and BNG

The broad spectrum of life on earth, which includes all species of flora and animals, fungi, microbes as well as the ecosystems they produce, is referred to as biodiversity. It incorporates genetic variation between species in addition to diversity within ecosystems, as a result, supporting vital services such as food, medicine, climate control and the filtering of air and water. (National Geographic, 2024) Ecosystems rely on biodiversity in order to maintain good health and supply a range of vital materials and amenities to mankind. (Global Citizen, 2023)

Vital services provided by ecosystems are in constant danger due to the decrease of biodiversity, originating from human-induced factors including pollution, habitat loss and global warming. Thus, resulting in irritation within ecological equilibrium. Therefore species, ecosystems and processes that sustain life could perish. (Global Citizen, 2023)

Biodiversity Net Gain (BNG) is a process concentrated on the formation and enrichment of natural habitats, ensuring that development projects result in positive and assessable outcomes on biodiversity, above the predeveloped environment. (Gov.uk, 2024) The focus on habitat improvement and restoration is to ensure that development has a beneficial impact on biodiversity as a whole. This emphasises on the enhancement and restoration of ecosystems in order to ensure that development has a beneficial impact on biodiversity. (Global Citizen, 2023)

BNG Scoring is a way of measuring the ecological impact of development projects, guaranteeing that biodiversity is enhanced post-development. It uses biodiversity units calculated based on factors like habitat size, condition, uniqueness, and significance (Defra, 2024). The process compares pre-development and post-development scores, requiring at least a 10% net gain in biodiversity through habitat creation or restoration (CIEEM, 2023). Tools like Defra's Statutory Biodiversity Metric regulate these calculations, confirming consistent and visible assessments (Defra, 2024)

This project integrates AI-driven tools to support BNG scoring by automating the identification of flora and assessing habitat characteristics. By using tools like image processing and generative AI, the developed prototype enables accurate biodiversity unit calculations in line with the Defra Statutory Biodiversity Metric, not only updates compliance with BNG requirements but also improves decision-making for sustainable development and conservation efforts.

1.1.2 Generative AI and Image Processing

The term 'generative artificial intelligence' (AI) describes algorithms that have the capability of utilising patterns found within existing data in order to produce new material, including text, image and audio. Generative AI aids ecosystem modelling, foresight of species distribution, as well as the analysis of complex ecological data within biodiversity matters. Development, assessment and interpretation of graphic data via images are all aspects of image processing methods. Its potential application in biodiversity includes monitoring changes within habitats, analysing the health of ecosystems, streamlining the detection as well as recording flora and organisms of all kinds. (ITU AI for Good, 2023)

Image processing hold an important role in BNG by enabling the automated identification and analysis of plant species and habitats, which can extract detailed information from images, such as species type, density, and distribution, which are critical factors in calculating biodiversity (Defra, 2024). For this project, image processing serves as the foundation for precise flora detection, guaranteeing accurate and scalable data collection. By integrating image processing with AI tools, the prototype can assist efficient biodiversity assessments, aligning with BNG goals to achieve measurable improvements in ecosystem health while supporting informed decision-making in conservation and development planning.

1.1.3 AI's significance in conservation initiatives

Artificial Intelligence has emerged as a key instrument in conservation, establishing innovative methods in order to monitor and conserve biodiversity. AI-powered statistical data analysis aids in maintaining an eye on endangered organisms. AI driven solutions provide prediction models for biodiversity patterns and habitat visualisation via data collection automation, classification of species and ecological data interpretation. (World Economic Forum, 2024)

1.1.4 Prompt Engineering and its significance

A key approach in the development and advancement of generative AI systems is known as prompt engineering, which involves thoughtful construction inputs in order to steer the AI model's responses in specific and accurate routes. This way, the system has the capability to comprehend complex requests and supply specific, context-aware results by personalising these prompts. (OpenAI, 2024)

Prompt engineering in the context of Biodiversity Net Gain (BNG) implies that the chatbot can precisely read user inputs and provide a straightforward and easy-to-understand description of the results of flora image-processing analysis. Serving as an efficient tool, making it possible to conveniently create, assess and refine code for the construction of BNG models. Through the creation of accurate prompts, developers can take advantage of multiple features of generative AI models such as Claude and GPT-4o in order to improve

biodiversity assessment methods as well as algorithms for analysing flora images. By ensuring that models are specifically modified for certain scenarios, prompt engineering can optimise consistency and adaptability.

In addition, prompt engineering is proactively essential to the user interface due to fact that it enables the chatbots to comprehend user input and deliver results in a human-readable way. For instance, the chat bot can have the ability to convert complex information into precise and straightforward descriptions, or even useful insights, when a user provides an image of flora for inspection. This unique advantage of prompt engineering ensures that the system remains practical and user-friendly for personnel such as environmentalists, legislators and other stakeholders while simultaneously enhancing the precision of BNG-related jobs.

Prompt engineering acts as the bridge that connects users to AI, making it easier to obtain efficient results. In the context of Biodiversity Net Gain (BNG), it helps guide AI tools to analyse habitats, identify plants, and generate BNG-related code, playing a key role in creating and refining the code is needed to build and test the AI chatbot prototype. By crafting clear prompts, it ensures that the AI can work efficiently with multiple models like GPT-4 or Claude, confirming the outputs align with BNG goals. This makes it simpler for users to interact with the system and get the data needed.

1.1.5 Existing Gaps in Tools and Methods

Despite gradual advancements, unmet necessities regarding technologies and approaches towards effective monitorisation and enhancement of Biodiversity Net Gain (BNG). Current strategies of assessing biodiversity are frequently lengthy and time-consuming, in addition to the lack of real-time data evaluation. To be able to provide reliable, accurate and adaptable methods for biodiversity monitoring and development, integrated systems which incorporate generative AI and image processing are essential. Reducing such gaps is crucial in order to attain sustainable development initiatives, as well as ensuring that they help the environment. (PBC Today – Digital Construction News, 2023).

1.1.6 Importance of AI in Addressing Biodiversity Challenges

Overcoming worldwide environmental challenges and achieving sustainability objectives require the development of AI-driven solutions for biodiversity observation. Natural resources which humans depend on for safe air, water and food are based on biodiversity. Considering the gradual reduction of biodiversity brought on by loss of habitats, global warming and deforestation, novel and feasible solutions are required. By producing precise, real-time data on the number of species as well as environmental welfare, the implementation of AI technologies such as generative models and image processing into monitoring diversity could entirely transform preservation efforts. (Global Citizen, 2023)

Such technologies provide practical insights to stakeholders, including biologists and legislators, permitting accurate decision-making for restoring ecosystems as well as maintaining them. Additionally, this project assists the UK's compliance

to the Kumming-Montreal Global Biodiversity Framework (JNCC, 2024), placing a strong emphasis on stimulating efficient use of terrestrial environments and reducing the loss of biodiversity. (House of Commons Library, 2024) This approach has the potential to improve ecological technology through the integration of AI, demonstrating the effectiveness of interdisciplinary approaches in tackling important global issues. In the end, this project is a step towards fusing discussion and progress, creating a future where these technologies support a sustainable society.

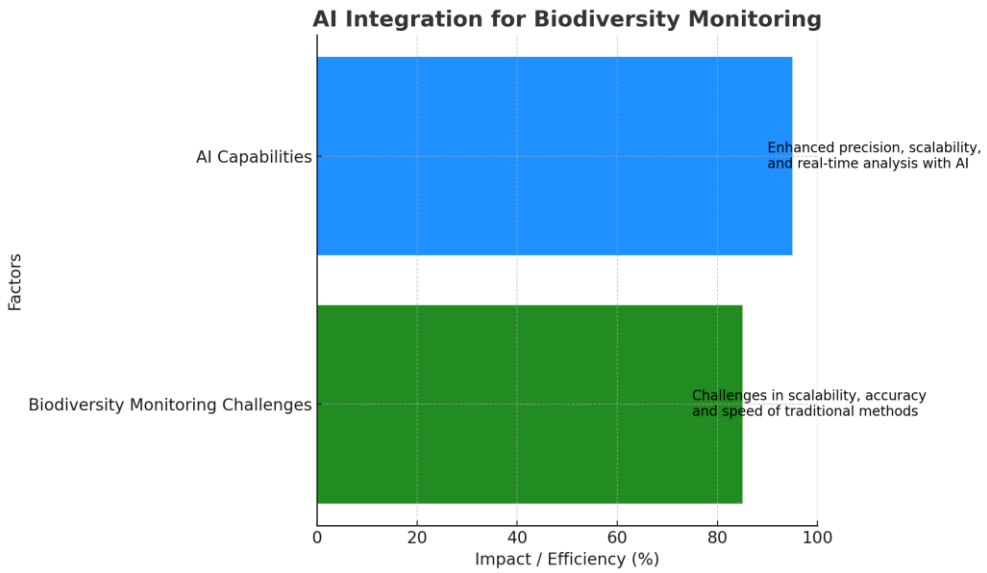


Figure 1: Comparison of Challenges and Capabilities in Biodiversity Monitoring

Figure 1 above highlights the effectiveness and impact of integrating AI into biodiversity monitoring in comparison to more conventional approaches. The difficulties encountered in conventional biodiversity monitoring are contrasted with the improved capabilities provided by AI, including scalability, accuracy, and real-time analysis.

1.1.7 Challenges in establishing and estimating BNG

Stakeholders face a range of challenges to genuinely realise the environmental benefits that Biodiversity Net Gain (BNG) promises. Lack of consistent criteria and procedures for measuring biodiversity outcomes leads to inconsistencies both within and across applications and evaluations of different programs. In order to achieve the anticipated environmental benefits from Biodiversity Net Gain (BNG), stakeholders have to overcome a number of high-profile challenges. An associated lack of common standards is reflected in inconsistent applications and appraisals of biodiversity impacts derived from diverse programs. This therefore may question the validity and success of BNG applications. (Biodiversity Oxford, 2023)

Also, the appropriate preparation, execution and administration of BNG efforts are hindered by the shortage of competent personnel experienced in BNG principles and methods. Such skill gaps may inhibit BNG operations from expanding and therefore may deliver inefficient results. (Surrey Wildlife Trust, 2023)

Financial limitations are a further concern due to the fact that it might be difficult to obtain adequate funding for the maintenance and analysis of biodiversity advances in the long-term. Lacking stable investment, BNG initiatives' feasibility and solidity are at risk. (The Times, 2024)

Furthermore, it may be challenging to embed BNG into traditional developmental and strategic applications, needing extensive alterations within regulations and procedures. Delays, as well as gradual additional costs could arise from this integration process, which may discourage stakeholders from providing BNG programs their mere focus. (Natural England Blog, 2023)

In order to overcome such challenges, personnel such as legislators, developers, ecologists and financial institutions must collaborate to formulate precise policies, improve capacity, obtain finance as well as clarify procedures for the effective deployment of BNG.

1.2 Project aims and objectives

The main aim of this project is to create a generative AI-powered chatbot prototype to create code that can support BNG initiatives through automatic image processing-based flora recognition and analysis. Expanding on this, the objective is to ensure the chatbot holds accurate and scalable biodiversity monitoring with the intention to close the gap between the latest AI technology and ecological protection. For increased interaction with users as well as offer insightful data, the chatbot will make us of forefront generative AI models such as GPT-4o.

The particular objectives consist of:

1. Exploration of efficient AI chatbot-based prompt engineering for code development and validation that can recognize user inputs with ease along with providing insightful, straightforward results through the analysis of flora images.
2. Employing prompt engineering in order to streamline the development, testing and evaluation of the underlying chatbot's model in addition to its interactions with users.
3. Using AI image processing algorithms in order to allow biodiversity analyses within a wide variety of conditions via the identification and recognition of plant varieties from provided images.
4. Making use of transfer learning to alter current plant-identifier services with the use of their API, in this case, Pl@ntNet's flora identification tool, alongside the use of datasets such as Oxford LibGuides' Plant Taxonomy database for increased species recognition precision.
5. Assessing the prototype's accuracy and dependability through testing in order to view its effectiveness in real-world conversation scenarios.

   Overall, this project intends to create a novel tool that aids biodiversity conservation efforts by combining generative AI, prompt engineering, transfer learning and already- existing useful resources.

1.3 Methodology

The methodology for this project focuses on creating an AI-driven chatbot designed to assist in Biodiversity Net Gain (BNG) assessments, operating as a code-generation tool to produce statistics based on image-processing outputs. First, datasets of flora images, sourced from taxonomic databases, will be pre-processed to train the AI model. Pretrained image-processing models will be fine-tuned for accurate flora detection and habitat analysis specific to BNG scoring criteria. Prompt engineering is the core of the chatbot's functionality. Specific prompts will guide generative AI systems, such as GPT-4, to produce Python scripts that analyse biodiversity data. These scripts will include functions to calculate habitat areas, assess condition scores, and generate biodiversity units. By using generative AI models, the chatbot ensures robust and flexible code output, enabling users to streamline BNG assessments efficiently.

To integrate BNG scoring, the prototype incorporates the Defra Biodiversity Metric by embedding its scoring criteria—such as habitat distinctiveness, condition, and area, into the code-generation pipeline. Validation and testing is also ensured. Flora detection accuracy is measured using performance metrics like precision and recall, while AI-generated BNG scores are cross-referenced with manual calculations to ensure consistency and compliance with established standards. This approach ensures the reliability and applicability of the chatbot in real-world conservation scenarios.

1.4 Report structure

The rest of the report is organised as follows: **Chapter 2** provides a literature review of relevant technologies, focusing on generative AI and applications relevant to biodiversity, and the use of image-processing models for flora detection. It also explores existing BNG assessment tools and methods. **Chapter 3** outlines the project planning and methodology, explaining the integration of prompt engineering and generative AI into the chatbot's development process. **Chapter 4** presents the proposed methodology workflow, including a detailed explanation of the data collection, preprocessing steps, and integration of BNG scoring metrics. This chapter also discusses the experiments conducted to refine and evaluate the chatbot. **Chapter 5** focuses on the evaluation process and the results obtained from testing the chatbot's performance in producing BNG statistics. Finally, **Chapter 6** concludes the report by summarising the project's achievements and discussing future work, including potential advancements in biodiversity monitoring and AI-driven conservation efforts.

2. Chapter 2: Literature review

The following literature review explores the foundational concepts and technologies that underpin this project, focusing on their relevance to Biodiversity Net Gain (BNG) assessment and conservation efforts. It begins by examining the principles of BNG, highlighting the importance of biodiversity monitoring in achieving environmental sustainability. Key technologies, including image processing and generative AI, are analysed to understand their applications in flora detection and habitat assessment. The review also investigates the role of machine learning and deep learning architectures in automating biodiversity analysis, with a focus on their integration into BNG scoring systems. Additionally, existing tools and methodologies which could be relevant to biodiversity conservation are critically reviewed to identify gaps and opportunities for improvement. This chapter lays the groundwork for understanding how prompt engineering, AI models as well as other existing tool can be used to address the challenges of BNG assessments and improve conservation practices.

2.1 Related Technologies

2.1.1 PictureThis – Plant Identifier App

PictureThis is a popular plant identification software that lets users identify different kinds of plants just by taking a picture of them. The app claims to have an accuracy of over 98% in identifying over 400,000 plant species worldwide (Google Play, 2024). The application can identify a broad variety of species since its AI model has been trained on a large dataset that includes millions of plant images. The training contributes to the model's excellent accuracy rate by enabling it to generalise across various plant species and scenarios (Android Central Forums, 2018) The app's web user interface is shown in the figures below:

Figure 2: *PictureThis interface (Screenshots from picturethisai.com)*

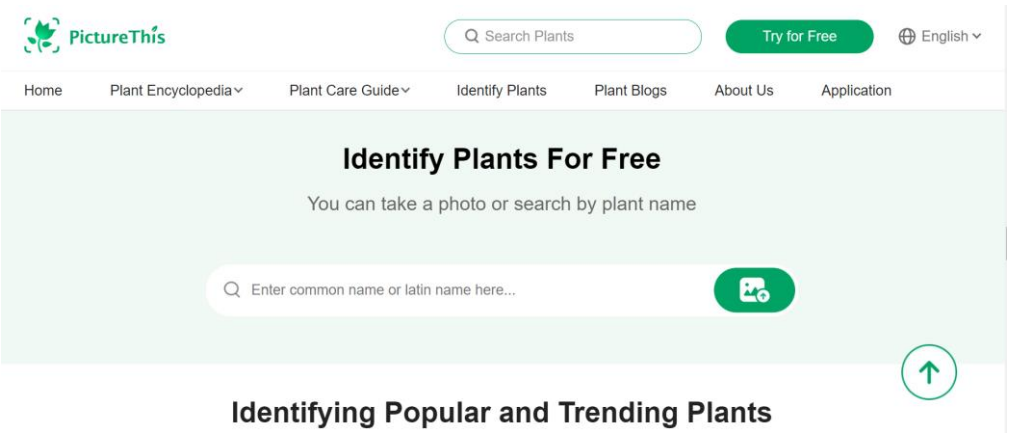


Figure 2.1: Home

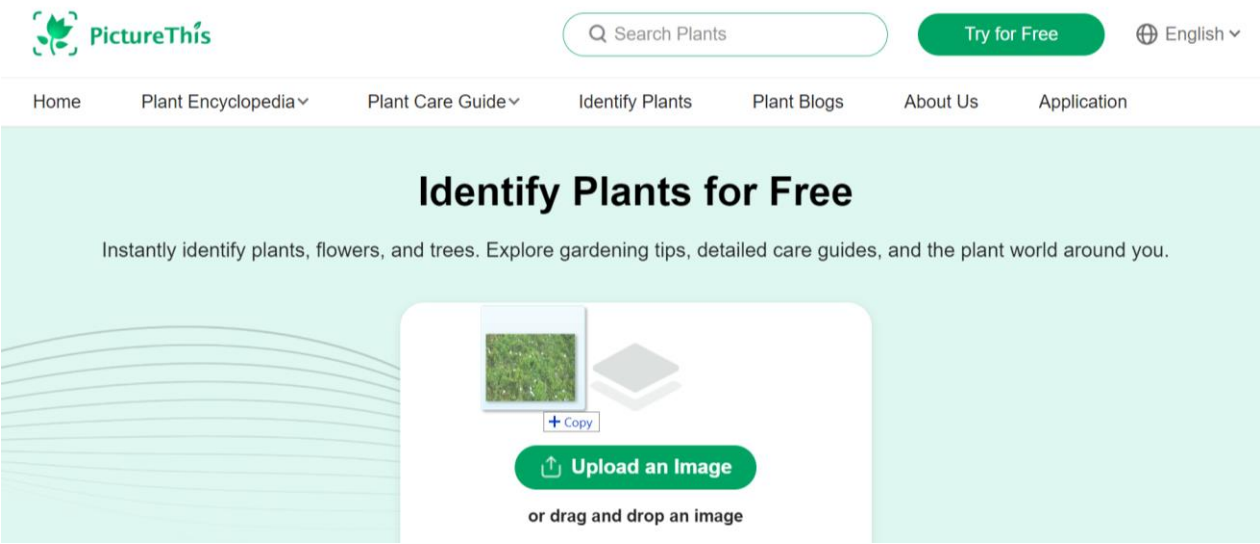


Figure 2.2: Uploaded image being used

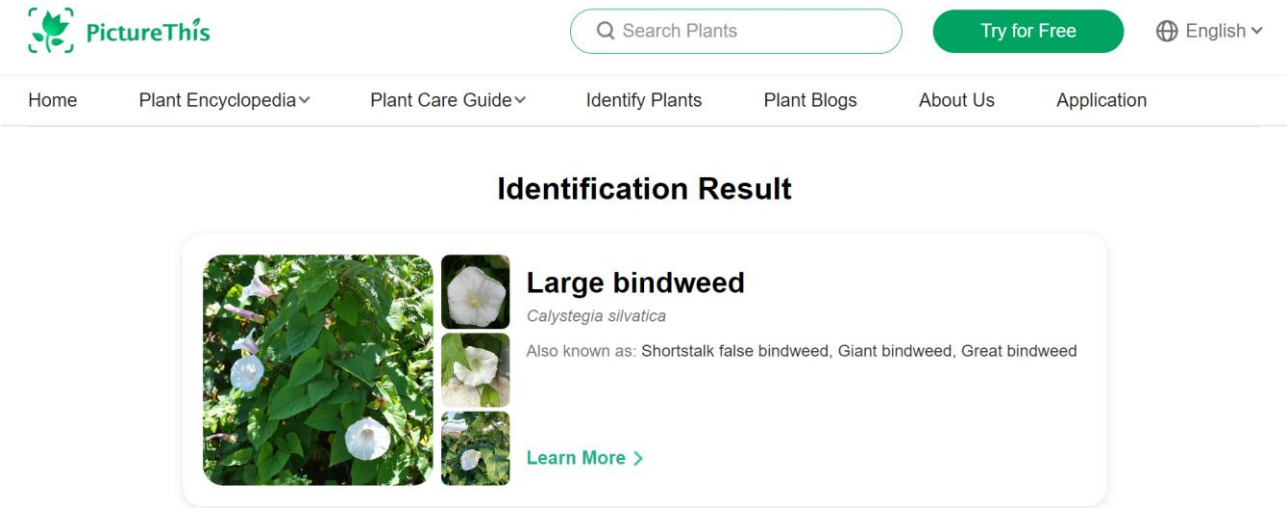


Figure 2.3: The results

As displayed above, a user is able to interact with PictureThis by taking a photo of a plant (if using the app), or as shown, upload an existing image. The app then processes the image through its AI engine to identify the plant species. As well as identification, PictureThis offers extra features such as plant care guides, problem diagnosis for sick plants, and a comprehensive plant encyclopaedia (PictureThisAI, 2024).

The core functionality of PictureThis relies on advanced AI and deep learning technologies. By using convolutional neural networks (CNNs) and a class of deep learning algorithms particularly effective in image recognition tasks, the app analyses user-submitted photos to detect and classify plant species based on learned features such as leaf shape, colour, and texture. This approach allows the app to provide instantaneous and accurate plant identification (Android Central Forums, 2018). As a result, the combination of AI and deep learning technologies enables PictureThis to deliver a user-friendly and efficient plant identification experience.

2.1.2 Biodiversity Bot - YesChat

Biodiversity Bot, accessible via YesChat, is an AI-powered tool designed to provide detailed information and insights on biodiversity, supporting both educational and research purposes. The bot has the ability to process large volumes of data from various sources to identify patterns, track changes in habitats, and monitor species populations, offering powerful tools for assessing biodiversity (SmartSocs, 2024). Below are screenshots of the bot represented as labelled figures:

Figure 3: *Biodiversity Bot Interface (Screenshots from yeschat.ai)*

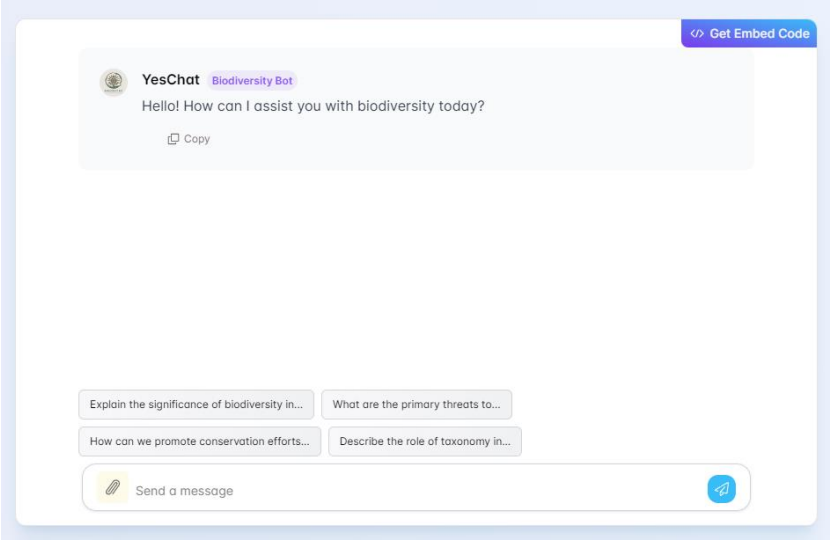


Figure 3.1: Start page

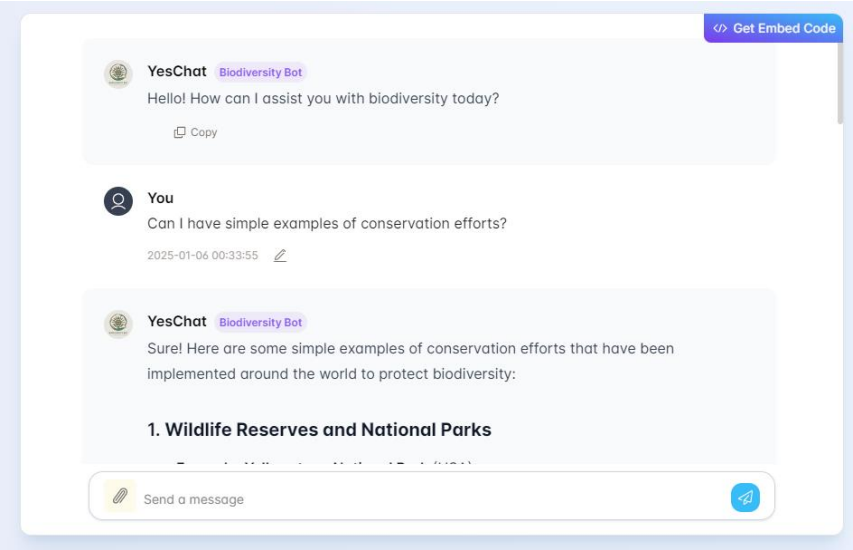


Figure 3.2: Displayed detailed answer after simple input

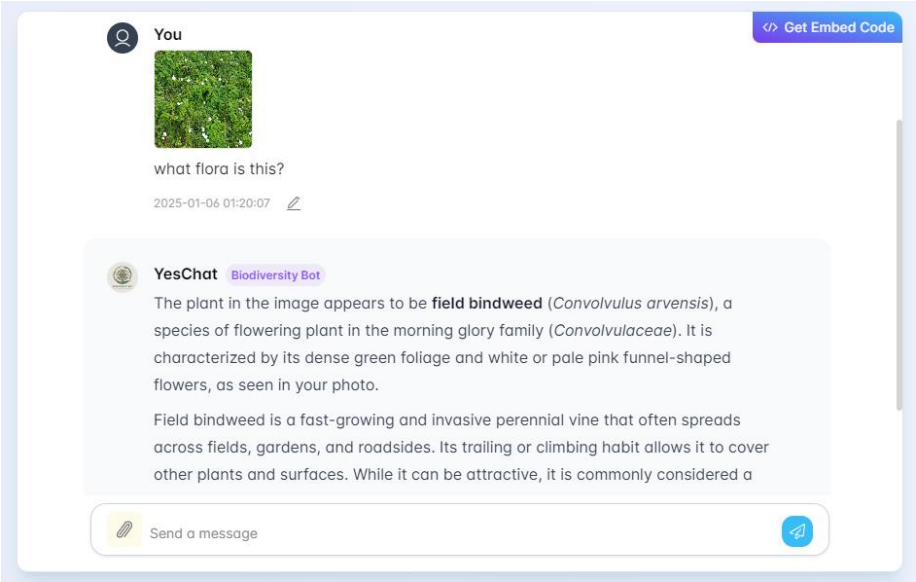


Figure 3.2: ID output after image input

The Biodiversity Bot is built on the foundation of OpenAI's GPT-4o, a natural language processing (NLP) model which enables the bot to understand user queries with the aid of advanced language understanding capabilities. GPT-4o interprets natural language inputs, including ambiguous or incomplete descriptions. The model generates human-like, contextually relevant, and detailed responses based on the input provided (OpenAI, 2023).

Users can upload photos of plants, which are analysed to extract features based on patterns which are then compared against a vast database of plant species, enabling accurate identification. GPT-4o's multimodal capabilities allow it to handle both textual and visual inputs. The image and text processing modules work together to enhance identification accuracy. Therefore, users are able to provide both a photo and descriptive keywords at the same time, as portrayed in figure 3.2.

The Biodiversity Bot is continuously updated via databases such as the Global Biodiversity Information Facility (GBIF) database: A repository of plant species distributions and classifications, and the PlantNet Database: Offers datasets for plant taxonomy and identification, contributing to accurate species recognition.

By combining the language-processing power of GPT-4 with image recognition and specialised datasets, the Biodiversity Bot delivers an ideal solution for plant identification and biodiversity education.

2.1.3 Pl@ntNet identification tool

PlantNet is an application that uses machine learning techniques to automatically identify plants from user-submitted photos. The initiative, which was created by a group of French research organisations, including CIRAD, INRAE, INRIA, and IRD, is to help experts and beginners alike correctly identify plant species. The following screenshots represents as figures show the user interface and the app's functionalities:

Figure 4: *Pl@ntNet Interface (Screenshots from identify.plantnet.org)*

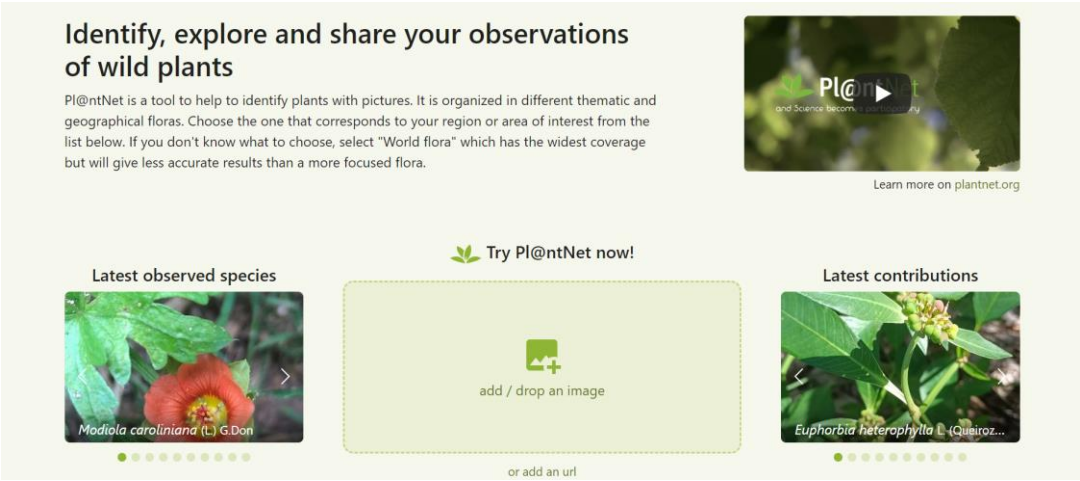


Figure 4.1: Start page

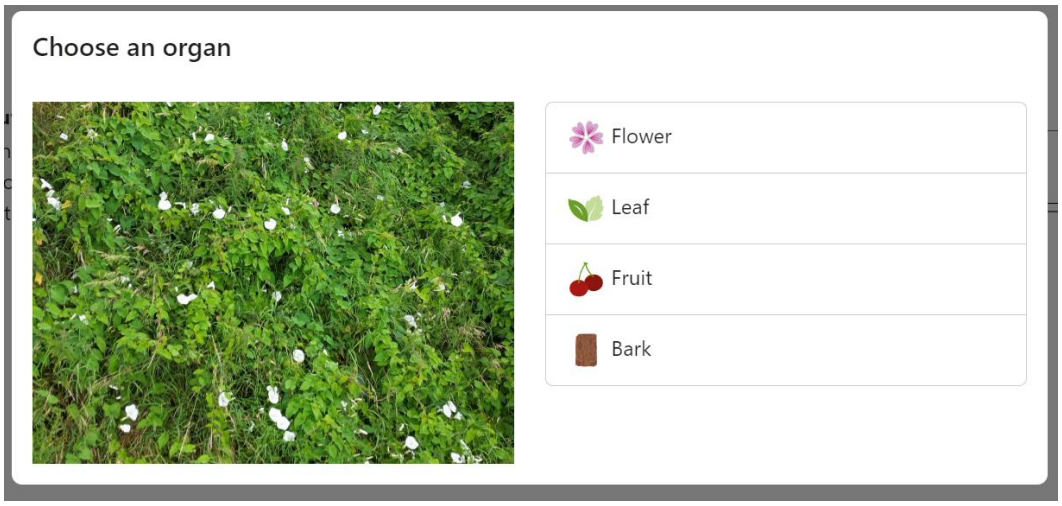


Figure 4.2: Organ selection page

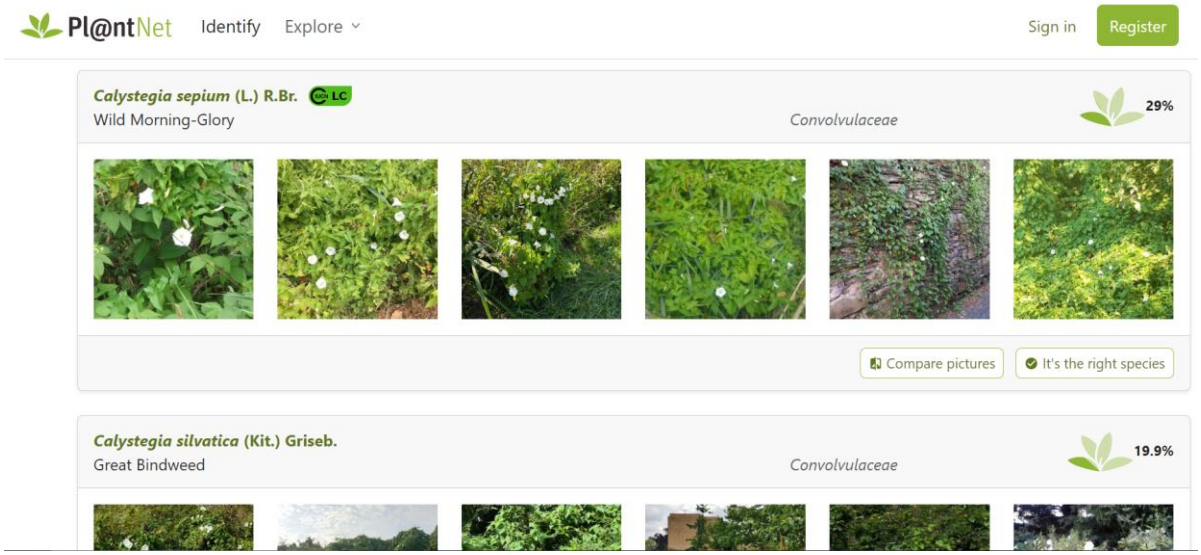


Figure 4.3: Generated results page

The platform employs deep learning algorithms to analyse images and identify plant species, scanning the imported photos across an extensive dataset in order to compare similarities with existing images of the matched flora. Integrated models are trained on extensive datasets involving millions of plant images, enabling the system to recognise a wide variety of species based on structural features such as leaves, flowers, fruits, and stems (GBIF 2023).

The 300K dataset from Pl@ntNet, which comprises more than 306,000 photos with 1,081 species, is classified by a long-tailed distribution and significant label ambiguity, which reflects the difficulties of plant identification in the real world where certain species are undersold. The structure of the dataset maintains the ambiguity that exists among species that belong to the same class, which facilitates the creation of models that can manage such complexity (Pl@ntNet, 2023).

Users can identify plants in a specific setting, making it a useful tool for research and teaching. With its vast database and easy-to-use interface, the platform has become an invaluable tool for ecological education, biodiversity research, and conservation initiatives. Pl@ntNet facilitates tracking the distribution of plant species and the identification of ecological changes over time by allowing for the large-scale collecting of data through user contributions (Pl@ntNet, 2023).

2.2 *Comparison of existing technologies*

The Biodiversity Bot, PictureThis, and Pl@ntNet applications all aim to facilitate plant identification and raise ecological consciousness. Convolutional neural networks (CNNs), in particular, are AI-driven algorithms used by all three for species identification and picture recognition. Every platform is designed to be easily navigable by users, assisting researchers and enthusiasts in activities pertaining to biodiversity. Furthermore, they use vast data sets to train their models. PictureThis provides curated datasets for refined findings, whereas Pl@ntNet's database, which has millions of user-submitted photographs, focusses on citizen research. With the help of GPT-4, Biodiversity Bot offers textual and image-based identification by fusing detailed species knowledge with visual recognition.

Even while these systems are similar, their methods and features are very different. Pl@ntNet is a community-driven, open-source platform that focusses on monitoring biodiversity and provides comprehensive coverage of plant species with high ecological importance. The PictureThis app, however, is commercial and prioritises user-friendliness and easy design for casual users, where subscriptions are frequently needed for more complex functions. Because Biodiversity Bot incorporates generative AI, it is unique in that it allows for conversational, in-depth interactions and species insights that go beyond identification to include biological context, habitat data, and conservation status. Similarities and differences of the three tools are represented below in the following table, (Table 1).

Table 1: Existing technologies comparison

Feature	Pl@ntNet	PictureThis	Biodiversity Bot
Goal	Community-driven biodiversity studies	Casual plant identification	Conversational biodiversity insights
AI technology	CNNs for image recognition	CNNs for image recognition	GPT-4 for text and image processing
Dataset	Open-source, user-contributed	Curated commercial datasets	External datasets (e.g., GBIF, PlantNet)
Species coverage	Extensive, including rare species	Common plants, less focus on rare species	Extensive with ecological insights
User Interface	Scientific and detailed	User-friendly, commercial focus	Conversational and educational
Cost	Free	Subscription-based	Free
Additional features	Biodiversity monitoring, citizen science	Plant care tips	Conservation status, habitat details